Scale Exercises - Section IV, Level 31

The pentatonic section still starts on A.

Major Pentatonic Two-Octave Six-String Neck Ladder: A to A

Extend the exact Level 30 shape through all six strings before moving the complete frame one fret.

1. Extend the exact frame

Root fret R: E R,R+2; A and D R-1,R+2; G R-1,R+1; B R,R+2; high e R.
Fingers: 2-4 | 1-4 | 1-4 | 1-3 | 2-4 | 2.

2. Cross all six strings

The ascent uses **2-2-2-2-2-1 notes per string**. Rebuild the G-to-B geometry instead of copying one string pattern upward.

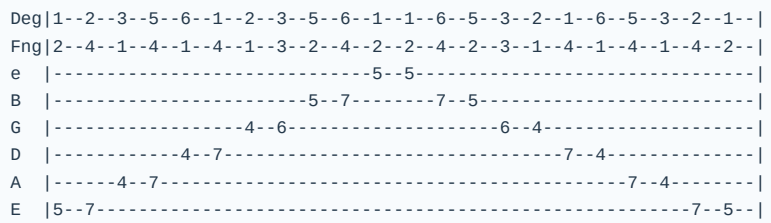
3. Preserve rhythm and pick cycle

Repeat the high root at the turn. The full up/down cell has an even **22 notes**, so each new key begins with the same pick orientation.

Worked example - A major pentatonic at low-E fret 5

Scale notes: A B C# E F# | A B C# E F# A. The high root is repeated at the turn.

LOW ROOT -> MIDDLE ROOT -> HIGH ROOT | HIGH ROOT -> MIDDLE ROOT -> LOW ROOT



KEY ROUTE A -> Bb -> B -> C -> Db -> D -> Eb -> E -> F -> Gb -> G -> Ab -> A, then reverse.

AUDIT Hear all three roots evenly; keep the B-string crossing and one-fret relocation invisible.

Session protocol

1. Master the A-major-pentatonic 22-note cell before adding a shift.
2. Start 10-15 BPM below your clean Level 30 tempo.
3. Use 2-3 neighboring rungs and pause one beat after each cell.
4. Repeat with a downstroke start and an upstroke start.
5. Remove the pause; connect the route and reverse at fret 17.
6. Add 2 BPM after 3 clean round trips; reduce 4-6 BPM after 2 errors.

Level 31 - Chromatic Major-Pentatonic Route

Ascending: complete one 22-note cell, then move the frame +1 fret

Descending: reverse the cards and move -1 fret

Rung	Key	Low-E root	Frame	Scale tones (degrees 1, 2, 3, 5, 6)
01	A major pentatonic	fret 5	4-7	A B C# E F#
02	Bb major pentatonic	fret 6	5-8	Bb C D F G
03	B major pentatonic	fret 7	6-9	B C# D# F# G#
04	C major pentatonic	fret 8	7-10	C D E G A
05	Db major pentatonic	fret 9	8-11	Db Eb F Ab Bb
06	D major pentatonic	fret 10	9-12	D E F# A B
07	Eb major pentatonic	fret 11	10-13	Eb F G Bb C
08	E major pentatonic	fret 12	11-14	E F# G# B C#
09	F major pentatonic	fret 13	12-15	F G A C D
10	Gb major pentatonic	fret 14	13-16	Gb Ab Bb Db Eb
11	G major pentatonic	fret 15	14-17	G A B D E
12	Ab major pentatonic	fret 16	15-18	Ab Bb C Eb F
13	A major pentatonic - octave	fret 17	16-19	A B C# E F#

How to use the map

Up pass:

rungs 01-13.

Turn:

complete the high-A cell at fret 17, then reverse.

Down pass:

rungs 12-01. Skip the high-A card unless starting a new round trip.

Key names:

say the printed major key before playing its low root.

Goal:

preserve one five-note formula while both octaves rise and fall chromatically.

Mastery gate before the next major-pentatonic level

Name every key before playing it; say 1-2-3-5-6 through both octaves; keep the four-fret frame compact; maintain strict alternate picking across all six strings; make the low, middle, and high roots sound intentional; land every one-fret relocation without sliding; and complete the fret-5 to fret-17 to fret-5 route with both pick starts on two separate days.

ROUND-TRIP COUNT 25 cells x 22 notes = 550 picked notes.

Level 31 - Major Pentatonic Two-Octave Six-String Neck Ladder

Rungs 01-03: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

01 A MAJOR PENTATONIC

ROOT E5 | FRAME 4-7

Notes: A B C# E F# A | B C# E F# A

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------5--5--------------------------------|
B  |----------------------5--7--------7--5----------------------------|
G  |------------------4--6--------------------6--4--------------------|
D  |------------4--7---------------------------------7--4-------------|
A  |------4--7-------------------------------------------7--4--------|
E  |5--7-------------------------------------------------------7--5--|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

02 BB MAJOR PENTATONIC

ROOT E6 | FRAME 5-8

Notes: Bb C D F G Bb | C D F G Bb

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------6--6--------------------------------|
B  |----------------------6--8--------8--6----------------------------|
G  |------------------5--7--------------------7--5--------------------|
D  |------------5--8---------------------------------8--5-------------|
A  |------5--8-------------------------------------------8--5--------|
E  |6--8-------------------------------------------------------8--6--|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

03 B MAJOR PENTATONIC

ROOT E7 | FRAME 6-9

Notes: B C# D# F# G# B | C# D# F# G# B

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------7--7--------------------------------|
B  |----------------------7--9--------9--7----------------------------|
G  |------------------6--8--------------------8--6--------------------|
D  |------------6--9---------------------------------9--6-------------|
A  |------6--9-------------------------------------------9--6--------|
E  |7--9-------------------------------------------------------9--7--|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 31 - Major Pentatonic Two-Octave Six-String Neck Ladder

Rungs 04-06: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

04 C MAJOR PENTATONIC

ROOT E8 | FRAME 7-10

Notes: C D E G A C | D E G A C

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------8--8-------------------------------|
B  |----------------------8--10-------10-8----------------------------|
G  |------------------7--9--------------------9--7--------------------|
D  |------------7--10-------------------------------10-7--------------|
A  |------7--10-----------------------------------------10-7---------|
E  |8--10-------------------------------------------------------10-8--|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

05 DB MAJOR PENTATONIC

ROOT E9 | FRAME 8-11

Notes: Db Eb F Ab Bb Db | Eb F Ab Bb Db

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------9--9-------------------------------|
B  |----------------------9--11-------11-9----------------------------|
G  |------------------8--10--------------------10-8--------------------|
D  |------------8--11-------------------------------11-8--------------|
A  |------8--11-----------------------------------------11-8---------|
E  |9--11-------------------------------------------------------11-9--|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

06 D MAJOR PENTATONIC

ROOT E10 | FRAME 9-12

Notes: D E F# A B D | E F# A B D

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------10-10-----------------------------|
B  |----------------------10-12-------12-10----------------------------|
G  |------------------9--11--------------------11-9--------------------|
D  |------------9--12-------------------------------12-9--------------|
A  |------9--12-----------------------------------------12-9---------|
E  |10-12-------------------------------------------------------12-10-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 31 - Major Pentatonic Two-Octave Six-String Neck Ladder

Rungs 07-09: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

07 EB MAJOR PENTATONIC

ROOT E11 | FRAME 10-13

Notes: Eb F G Bb C Eb | F G Bb C Eb

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------11-11-------------------------------|
B  |-----------------------11-13-------13-11-----------------------------|
G  |-------------------10-12---------------------12-10-------------------|
D  |------------10-13-------------------------------13-10--------------|
A  |------10-13-------------------------------------13-10-------|
E  |11-13-------------------------------------------------13-11-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

08 E MAJOR PENTATONIC

ROOT E12 | FRAME 11-14

Notes: E F# G# B C# E | F# G# B C# E

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------12-12-------------------------------|
B  |-----------------------12-14-------14-12-----------------------------|
G  |-------------------11-13---------------------13-11-------------------|
D  |------------11-14-------------------------------14-11--------------|
A  |------11-14-------------------------------------14-11-------|
E  |12-14-------------------------------------------------14-12-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

09 F MAJOR PENTATONIC

ROOT E13 | FRAME 12-15

Notes: F G A C D F | G A C D F

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------13-13-------------------------------|
B  |-----------------------13-15-------15-13-----------------------------|
G  |-------------------12-14---------------------14-12-------------------|
D  |------------12-15-------------------------------15-12--------------|
A  |------12-15-------------------------------------15-12-------|
E  |13-15-------------------------------------------------15-13-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 31 - Major Pentatonic Two-Octave Six-String Neck Ladder

Rungs 10-12: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

10 GB MAJOR PENTATONIC

ROOT E14 | FRAME 13-16

Notes: Gb Ab Bb Db Eb Gb | Ab Bb Db Eb Gb

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------14-14-------------------------------|
B  |-----------------------14-16-------16-14-----------------------------|
G  |-------------------13-15-----------------15-13-------------------|
D  |------------13-16-------------------------------16-13-------------|
A  |------13-16-----------------------------------------16-13-------|
E  |14-16-------------------------------------------------------16-14-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

11 G MAJOR PENTATONIC

ROOT E15 | FRAME 14-17

Notes: G A B D E G | A B D E G

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------15-15-------------------------------|
B  |-----------------------15-17-------17-15-----------------------------|
G  |-------------------14-16-------------------16-14-------------------|
D  |------------14-17-------------------------------17-14-------------|
A  |------14-17-----------------------------------------17-14-------|
E  |15-17-------------------------------------------------------17-15-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

12 AB MAJOR PENTATONIC

ROOT E16 | FRAME 15-18

Notes: Ab Bb C Eb F Ab | Bb C Eb F Ab

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-------------------------------16-16-------------------------------|
B  |-----------------------16-18-------18-16-----------------------------|
G  |-------------------15-17-------------------17-15-------------------|
D  |------------15-18-------------------------------18-15-------------|
A  |------15-18-----------------------------------------18-15-------|
E  |16-18-------------------------------------------------------18-16-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 31 - Major Pentatonic Two-Octave Six-String Neck Ladder

Rung 13 and full-ladder consolidation

For the descending pass, return through rungs 12-01 and move -1 fret

13 A MAJOR PENTATONIC - OCTAVE

ROOT E17 | FRAME 16-19

Notes: A B C# E F# A | B C# E F# A

22-NOTE TWO-OCTAVE CELL

```
Deg|1--2--3--5--6--1--2--3--5--6--1--1--6--5--3--2--1--6--5--3--2--1--|
Fng|2--4--1--4--1--4--1--3--2--4--2--2--4--2--3--1--4--1--4--1--4--2--|
e  |-----------------------------17-17-------------------------------|
B  |-----------------------17-19-------19-17-----------------------|
G  |-----------------16-18-------------------18-16-----------------|
D  |------------16-19-------------------------------19-16----------|
A  |------16-19-----------------------------------------19-16------|
E  |17-19-------------------------------------------------------19-17-|
```

Return to the low root; release pressure; move the entire frame one fret.

D-start and U-start

Full round-trip challenge

1. Start with A major pentatonic at low-E fret 5.
2. Play all 13 cells without restarting the pick cycle.
3. Reverse after the fret-17 cell and return through rungs 12-01.
4. Say each key name and sing 1-2-3-5-6 through both octaves.
5. Repeat with the opposite starting stroke.

Clean-round standard:

no open strings, missing pentatonic tone, unplanned doubled note, dragged shift, crossing hesitation, or volume drop at the middle or high root.

Advance: complete the route cleanly with both pick starts on two separate days.

550 picked notes per complete round trip.

After the ladder is clean

Root accents: accent only the low, middle, and high roots.
Chord-tone focus: accent 1, 3, and 5; hear 2 and 6 as color tones.
String-entry audit: accent the first note on every new string, then remove the accents.
Direction: begin selected sessions at fret 17 and descend first.
Random entry: choose any root fret from 5-17, name the key, and continue either way.

Practice log

Date	BPM	Start	Clean?	Notes

Next: Level 32 begins major-pentatonic sequence studies.